

DONGJIN KIM

Postdoctoral Associate

BioFrontiers Institute, Department of Chemical and Biological Engineering
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EDUCATION & AFFILIATION

2026 - present	Postdoctoral Associate with Prof. Saad Bhamla, BioFrontiers Institute, Department of Chemical and Biological Engineering, University of Colorado Boulder
2024 - 2026	Postdoctoral Associate, Department of Mechanical Engineering, Ajou University
2018 - 2024	Ph.D. with Prof. Je-Sung Koh, Department of Mechanical Engineering, Ajou University
2012 - 2018	B.Sc. Department of Mechanical Engineering, Ajou University

EXPERIENCES

May - June 2022	Visting Researcher with Prof. Saad Bhamla, Chemical and Biomolecular Engineering, Georgia Institute of Technology
Spring 2021	Lecturer for Dynamics, Department of Human Mechanical and Biomedical Engineering, Ewha Womans University

HONOURS AND AWARDS

2026	Sejong Science Fellowship (Overseas Track), National Research Foundation of Korea, <i>USD 52K</i>
2025	Top 10 Science & Technology News of the Year, Korean Federation of Science & Technology Societies
2025	Best Paper Award (Dean's Award) College of Engineering, Ajou University
2025	Best Paper Award, Korean Society of Mechanical Engineers Spring Conference
2024	Postdoctoral Domestic Fellowship, National Research Foundation of Korea, <i>USD 90K</i>
2023	Ph.D. Student Research Fellowship, National Research Foundation of Korea, <i>USD 15K</i>

PUBLICATIONS [\[GOOGLE SCHOLAR\]](#) * indicates co-authorship; † indicates corresponding author

Featured Publication

1. V. M. Ortega-Jimenez*†, **D. Kim***, S. Kumar, C. Kim, J.-S. Koh†, S. Bhamla†, Ultrafast elastocapillary fans control agile maneuvering in ripple bugs and robots. **Science** (2025, **Cover article**).
2. I. Hong*, D. Lim*, **D. Kim***, ..., J.-S. Koh†, S. Han†, and D. Kang†, Breathable, wearable skin analyzer for reliable long-term monitoring of skin barrier function and individual environmental health impacts. **Nature communications** (2025).
3. **D. Kim***, B. Kim*, B. Shin*, ..., J. S. Koh†, Actuating compact wearable augmented reality devices by multifunctional artificial muscle. **Nature Communications** (2022, **Featured in editor's highlight article**).
4. **D. Kim***, S. Han*, T. Kim, C. Kim, D. Lee, D. Kang†, J. S. Koh†, Design of a sensitive balloon sensor for safe human–robot interaction. **Sensors** (2021).
5. **D. Kim***, C. J. Park*, J.-S. Koh†, J. Ha†, Small-scale robots inspired by aquatic interfacial biolocomotion. **MRS Bulletin** (2024).
6. J. Ko, C. Kim, **D. Kim**, ..., J. S. Koh†, J. Cho†, High-performance electrified hydrogel actuators based on wrinkled nanomembrane electrodes for untethered insect-scale soft aquabots. **Science Robotics** (2022).
7. **D. Kim**, M. Gwon, B. Kim, V. M. Ortega-jimenez, S. Han, D. Kang, M. S. Bhamla, J. S. Koh†, Design of

- a Biologically Inspired Water-Walking Robot Powered by Artificial Muscle. **Micromachines** (2022).
8. M. Gwon, **D. Kim**, ..., J. S. Koh†, Scale dependence in hydrodynamic regime for jumping on water. **Nature Communications** (2023, **Featured in editor's highlight article**).
 9. Y. Roh*, M. Kim*, S. M. Won*, D. Lim, I. Hong, S. Lee, T. Kim, C. Kim, D. Lee, S. Im, G. Lee, **D. Kim**, ..., S. Han†, Vital signal sensing and manipulation of a microscale organ with a multifunctional soft gripper. **Science robotics** (2021).
 10. J. Ko, **D. Kim**, ..., J. S. Koh†, J. Cho†, Electroosmosis-Driven Hydrogel Actuators Using Hydrophobic/Hydrophilic Layer-By-Layer Assembly-Induced Crack Electrodes. **ACS Nano** (2020).

International Journal & Conference

11. C. Kim, D. Shin, I. Back, **D. Kim**, ..., J.-S. Koh†, Inflatable Robotics Arm Capable of Deploying and Retracting by Rolling for High-Packaging Ratio, **2024 IEEE RoboSoft** (2024)
12. T. Kim*, I. Hong*, Y. Roh*, **D. Kim**, ..., D. Kang†, Spider-inspired tunable mechanosensor for biomedical applications. **npj Flexible Electronics** (2023)
13. T. Kim*, I. Hong*, M. Kim*, S. Im*, Y. Roh, C. Kim, J. Lim, **D. Kim**, ..., D. Kang†, Ultra-stable and tough bioinspired crack-based tactile sensor for small legged robots. **npj Flexible Electronics** (2023)
14. B. Kim*, D. Lee*, **D. Kim**, S. Han, D. Kang, U. Kim†, J. Koh†, Control of Shape Memory Alloy Actuator via Electrostatic Capacitive Sensor for Meso-scale Mirror Tilting System, **2023 IEEE ICRA** (2023)
15. H. Yi, D. Kim, Y. Kim, **D. Kim**, J. S. Koh†, M. J. Kim†, 3D-printed attachable kinetic shading device with alternate actuation: Use of shape-memory alloy (SMA) for climate-adaptive responsive architecture. **Automation in Construction** (2020).
16. S. Chun, W. Son, C. Choi, H. Min, J. Kim, H. J. Lee, **D. Kim**, C. Kim, J. S. Koh, C. Pang†, Bioinspired Hairy Skin Electronics for Detecting the Direction and Incident Angle of Airflow. **ACS Applied Materials & Interfaces** (2019).

PATENTS

1. Shin B., Koh J., Lee H., **Kim D.**, Kim B., *Haptic glove apparatus and virtual reality apparatus including the same*. Filed Nov. 17, 2021 (US11,989,347 B2)
2. Shin B., Koh J., Lee H., **Kim D.**, Kim B., *Driving device including shape memory alloy*. Filed Sep. 17, 2021 (US11,661,927 B2)
3. Shin B., Koh J., **Kim D.**, Sung G., Lee H., *Image display device capable of multi-depth expression*. Filed Mar. 31, 2020 (US11,187,896 B2)
4. Kang D., Han S., Koh J., Seo S., Hong I., **Kim D.**, Lim D., *Wireless multi-sensor apparatus capable of stably measuring skin*. Filed Sep. 5, 2024 (WO2025/110431 A1)

PROJECTS AND PROPOSALS (PARTICIPATING RESEARCHER)

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| 2025 | Korea Health Industry Development Institute
<i>Wearable exosuit</i> (USD 1.45M, Project Contributor) |
| 2025 | Ministry of Trade, Industry and Energy, International Joint Research Project
<i>Intelligent soft robot</i> (USD 1.45M, Proposal Contributor) |
| 2024 | National Research Foundation of Korea, Global Research Laboratory Program
<i>Hyper actuating system</i> (USD 1.1M, Key researcher and Proposal Contributor) |

2023	Convergence Research Center for Tactile Standards <i>Integrated haptic system</i> (USD 22M, Proposal Contributor)
2022 - 2024	Korea Environmental Industry & Technology Institute <i>Skin monitoring device</i> (USD 1.3M, Project Contributor)
2019 - 2021	Samsung Advanced Institute of Technology <i>AR glasses with compact actuator</i> (USD 0.22M, Key researcher and Proposal Contributor)